07.

Profit & Loss

Type - 3

1. Heeru bought an old car for ₹47000 and spent ₹3000 on its repairing. If he sells the car for ₹58,000, then how much will be his profit %?

(a) 18% (b) 16%
(c) 17% (d) 15%

RRB RPF Constable -18/01/2019 (Shift-III)

Ans. (b) : Total cost price of the car = 47000 + 3000
= ₹ 50,000
Selling price of the car = ₹58,000
So, Profit = S.P. – C.P.
Profit = 58,000 – 50,000 = ₹ 8,000
Hence, Profit % = $\frac{8,000\times100}{50,000}$ = 16%

2. The cost price of 25 items is equal to the selling price of 20 items. Find the profit %.

(a) 50% (b) 25%
(c) 40% (d) 30%

RRB JE - 24/05/2019 (Shift-I)

Ans : (b) Profit % = $\left(\frac{25-20}{20}\right)\times100 = 25\%$

3. The profit on the sale of a product is 25%. How much will be the loss percent, if the values of the cost price and selling price are interchanged?

(a) Loss of 12% (b) Loss of 16%
(c) Loss of 20% (d) Loss of 25%

RRB JE - 27/05/2019 (Shift-II)

Ans : (c) Let the cost price = 100
Selling price = 100 + 25 = 125
According to the question,–
On, interchanging the value of cost price and selling price,
Cost price = 125
Selling price = 100
Loss % = $\frac{125-100}{125}\times100 = \frac{25}{125}\times100 = 20\%$

4. Two items whose cost price is ₹ 2500 each, one is sold at a profit of 5%. If the total profit is 20%, then find the profit on second item.

(a) 20% (b) 25%
(c) 30% (d) 35%

RRB JE - 01/06/2019 (Shift-III)

Ans. (d) Let the second item is sold on x% profit.
According to the question,,

$$2500\times\frac{(100+5)}{100}+2500\times\frac{(100+x)}{100}=5000\times\frac{(100+20)}{100}$$

25 × 105 + 25 (100 + x) = 50 × 120
2625 + 2500 + 25x = 6000
25x = 6000 – 5125
25x = 875
x = 35%

5. Kaveri bought a toy for ₹ 280 and sold it for ₹ 315. How much profit did she get?

(a) 17.5% (b) 12.5%
(c) 16% (d) 15.25%

RRB RPF SI -05/01/2019 (Shift-II)

Ans. (b) : Cost price of the toy = ₹280
Selling price of the toy = ₹315
Formula– Profit (p) = Selling price – Cost price
P = 315 –280 = ₹35
Formula– P% = $\frac{\text{Profit}}{\text{Cost Price}}\times100$
$= \frac{35}{280}\times100 = \frac{5\times100}{40} = \frac{50}{4} = 12.5\%$

6. Geeta prepares cakes for programs. She prepared a cake for birthday party and sold it for ₹ 700. The cost price for preparing the cake was ₹ 350. What is the profit % she gained?

(a) 200% (b) 50%
(c) 100% (d) 150%

RRB RPF Constable -17/01/2019 (Shift-I)

Ans. (c) : Cost price (CP) = ₹350
Selling price (SP) = ₹700
Profit % = $\frac{\text{SP - CP}}{\text{CP}}\times100$
$= \frac{700-350}{350}\times100 = \frac{350}{350}\times100 = 100\%$

7. Anurag gets a loss of 1/7 on cost price when sells a pen for ₹ 144. If the pen is sold for ₹ 189, then what is the profit %?

(a) 11% (b) 12.5%
(c) 11.5% (d) 14%

RRB Group-D – 17/09/2018 (Shift-I)

Ans : (b) Given–
Loss = = $\frac{\text{Cost Price}}{7}$
Selling price (SP) = 144
Hence, Loss = Cost price (CP) − Selling price (SP)
$\frac{CP}{7} = CP - 144$
$CP - \frac{CP}{7} = 144$
$\frac{6CP}{7} = 144$

CP = 24 × 7
CP = 168
New Selling price = 189

$$\text{Profit } \% = \frac{SP - CP}{CP} \times 100 = \frac{189 - 168}{168} \times 100$$

$$= \frac{21}{168} \times 100 = 12.50\%$$

8. If the cost price of 15 oranges is equal to the selling price of 20 oranges then loss % is:
(a) 30% (b) 75%
(c) 25% (d) 40%
RRB Group-D – 20/09/2018 (Shift-II)

Ans : (c) Let the cost price of an orange = ₹ 1.
Then the cost price of 20 oranges = ₹ 20
Cost price of 15 oranges = selling price of 20 oranges

$$\text{Then Loss } \% = \frac{20 - 15}{20} \times 100$$

Loss % = 25%

9. Shefali bought a cup-set for ₹ 575, but later she sold it for ₹ 506 due to being old. How much % loss did she get?
(a) 13% (b) 14%
(c) 16% (d) 12%
RRB Paramedical Exam – 20/07/2018 (Shift-I)

Ans : (d) The cost price of the cup-set = ₹ 575
Selling price = ₹ 506
Loss = Cost price – Selling price
= 575 – 506
Loss = ₹ 69

$$\text{Loss\%} = \frac{\text{Loss}}{\text{cost price}} \times 100 = \frac{69}{575} \times 100 = 12\%$$

10. A man buys 5 pens for ₹ 1 and sells 4 pens for ₹ 1. Find his profit %.
(a) 25% (b) 40%
(c) 50% (d) 20%
RRB Group-D – 24/09/2018 (Shift-II)

Ans : (a) The cost price of a pen $= ₹ \frac{1}{5}$

And the selling price of a pen $= ₹ \frac{1}{4}$

$$\text{Profit } = \frac{1}{4} - \frac{1}{5} = \frac{1}{20}$$

$$\text{Profit } \% = \frac{\frac{1}{20}}{\frac{1}{5}} \times 100$$

$$= \frac{5}{20} \times 100 = 5 \times 5 = 25\%$$

11. The selling price of an item with 16.5% profit was ₹ 466. If the item was sold for ₹ 330 then what % of loss would happen?
(a) 17.25 (b) 17.75
(c) 17 (d) 17.5
RRB Group-D – 15/10/2018 (Shift-I)

Ans : (d)

$$\text{Selling price} = \text{Cost price} \times \frac{(100 \pm \text{Profit/Loss})}{100}$$

$$466 = \text{Cost price } \frac{(100 + 16.5)}{100}$$

$$\text{Cost price } = \frac{466 \times 100}{116.5}$$

Cost price = 400
Loss = 400 – 330 = 70

$$\text{Loss\%} = \frac{70 \times 100}{400} = 17.5\%$$

12. The cost prices of two items are ₹ 200 and ₹ 600. A shopkeeper gets a profit of ₹ 300 by selling both. If the shopkeeper sells the first item at a 30% profit, then at which profit does he sell the second item?
(a) 50% (b) 30%
(c) 20% (d) 40%
RRB Group-D – 31/10/2018 (Shift-II)

Ans : (d) Profit on first item $= \frac{200 \times 30}{100} = 60$

$$\text{The profit on second item } = \frac{300 - 60}{600} \times 100$$

$$= \frac{240 \times 100}{600} = 40\%$$

13. A man sold two bicycles at a total profit of 20%. If he had bought them for ₹ 3500 each and sold the first one at a profit of 5%, then for how much profit (%) he must make on the other one?
(a) 20% (b) 35%
(c) 25% (d) 30%
RRB Group-D – 31/10/2018 (Shift-II)

Ans : (b) Total cost price = 3500 + 3500 = ₹ 7000

$$\text{Selling price at a profit of 20\%} = 7000 \times \frac{120}{100} = ₹\ 8400$$

Selling price of first cycle at a profit of 5%

$$= 3500 \times \frac{105}{100}$$

$S.P._I = 3675$
Rest = 8400 – 3675 = 4725
Hence, the selling price of second cycle = ₹ 4725
Profit = 4725 – 3500
= ₹ 1225

$$\text{Profit } \% = \frac{1225}{3500} \times 100$$

Profit % = 35%

14. A carpenter suffers a loss of 10% on selling a footboard for ₹ 72. How much profit (%) or loss will be if he sells the footboard for ₹ 96?
(a) Profit, 10% (b) Profit, 20%
(c) Loss, 16% (d) Loss, 25%
RRB Group-D – 04/10/2018 (Shift-II)

Ans : (b) Cost price $= \frac{\text{Selling Price}}{100 - \text{Loss}\%} \times 100$

$= \frac{72 \times 100}{100 - 10}$

$= \frac{7200}{90} = 80$

Again Selling price = ₹96

Profit% $= \frac{\text{Selling Price - Cost Price}}{\text{Cost Price}} \times 100$

$= \frac{96 - 80}{80} \times 100 = 20\%$

15. By selling an article for ₹ 979, a shopkeeper bears a loss of 11%. If the shopkeeper sells that article for ₹ 1232, then profit % is:

(a) 12% (b) 21%
(c) 11% (d) 14%

RRB NTPC 19.04.2016 Shift : 1

Ans : (a) $\frac{100 - \text{Loss}\%}{100 + \text{Profit}\%} = \frac{\text{First S.P.}}{\text{Second S.P.}}$

$\frac{100 - 11}{100 + x} = \frac{979}{1232}$ or $\frac{89}{100 + x} = \frac{979}{1232}$

100 + x = 112 or x = 12

∴ Profit % = 12%

16. Mr. Rajesh purchases a toy for ₹ 27.50 and sells it for ₹ 28.60 then profit % is:

(a) 5% (b) 4%
(c) 6% (d) 3%

RRB NTPC 17.01.2017 Shift-1

Ans : (b) Profit % $\frac{\text{Profit}}{\text{Cost Price}} \times 100$

$= \frac{28.60 - 27.50}{27.50} \times 100$

$= \frac{1.10 \times 100}{27.50} = \frac{110}{27.5} = 4\%$

17. 15 laptops were bought at a rate of ₹ 15,000 each and all of them were sold for ₹ 2,97,000. Find the profit %.

(a) 28% (b) 40%
(c) 33.335 (d) 32%

RRB NTPC 17.01.2017 Shift-2

Ans : (d) According to the question,,

Price of 15 laptops = 15000 × 15 = 225000

Profit % $= \frac{\text{Profit}}{\text{Cost Price}} \times 100$

Profit % $= \frac{297000 - 225000}{225000} \times 100$

$= \frac{72000}{225000} \times 100 = 32\%$

18. The selling price of an item is ₹ 1458 and its cost price is ₹ 1350. Find the profit %.

(a) 5% (b) 6%
(c) 7% (d) 8%

RRB NTPC 05.04.2016 Shift-1

Ans : (d) Given-

Selling price of the item = ₹1458

Cost price of the item = ₹1350

Profit% $= \frac{\text{Profit}}{\text{Cost Price}} \times 100 = \left(\frac{1458 - 1350}{1350}\right) \times 100$

$= \frac{108}{1350} \times 100 = 8\%$

19. A shopkeeper sells cricket bats in this way that the selling price of 35 bats is equal to the cost price of 50 bats. Find his profit %.

(a) 33.33% (b) 42.83%
(c) 50% (d) 60%

RRB NTPC 04.04.2016 Shift : 3

Ans : (b) If the selling price of a items is equal to the cost price of b items, then

Profit % $= \frac{b - a}{a} \times 100$

Hence, shopkeeper's profit % $= \frac{50 - 35}{35} \times 100$

$= \frac{15}{35} \times 100 = 42.83\%$

20. The selling price of an item inclusive of a 16% profit was ₹ 435. What would be the percentage loss if the item was sold for ₹ 330?

(a) 12.25 (b) 13
(c) 12 (d) 12.5

RRB ALP & Tec. (31-08-18 Shift-II)

Ans. (c) : The cost price of the item $= 435 \times \frac{100}{116} =$ ₹375

If the item was sold for ₹ 330 then,

Loss % $= \frac{375 - 330}{375} \times 100$

$= \frac{45}{375} \times 100 = 12\%$

21. By selling a table for ₹ 16,870, a shopkeeper suffers a loss of ₹ 1,080. His loss percentage (rounded off to one decimal place) is :

(a) 6.1% (b) 6.2%
(c) 6.4% (d) 6.0%

RRB ALP & Tec. (31-08-18 Shift-II)

Ans. (d): Selling price of the table = ₹16,870

Loss = ₹1080

Then, the cost price of the table

= 16870+1080 = ₹17,950

Loss % $= \frac{\text{Loss}}{\text{Cost Price}} \times 100 = \frac{1080}{17950} \times 100 = \frac{108000}{17950}$

= 6.0167

Loss % = 6.01%

22. Kaushik bought a toy for ₹ 160 and sold it for ₹ 180. The rate of profit was _____%

(a) 15.25 (b) 12.5
(c) 17.5 (d) 16

RRB ALP & Tec. (20-08-18 Shift-I)

Ans : (b) The cost price of the toy = 160
Selling price = 180
Profit % $= \frac{180-160}{160} \times 100 = \frac{1}{8} \times 100$
Profit % = 12.5%

23. The S.P of a washing machine is $1\frac{1}{3}$ of its C.P. Find the gain percent.
(a) 33% (b) 66%
(c) $33\frac{1}{3}\%$ (d) $66\frac{1}{3}\%$

RRB ALP & Tec. (20-08-18 Shift-II)

Ans : (c) Let Cost price = ₹100
Selling price = 100 of $1\frac{1}{3} = 100 \times \frac{4}{3} = \frac{400}{3}$
Profit $= \frac{400}{3} - 100 = \frac{100}{3}$
Profit% $= \frac{\text{Profit}}{\text{Cost Price}} \times 100 = \frac{\frac{100}{3}}{100} \times 100 = \frac{100}{3}\%$
$= 33\frac{1}{3}\%$

Type - 2

24. A person gets a profit equal to the cost price of 3 beds, by selling 18 beds for ₹ 16,800. Find the cost price of one bed.
(a) ₹750 (b) ₹650
(c) ₹800 (d) ₹700

RRB JE - 29/05/2019 (Shift-I)

Ans : (c) Let the cost price of a bed = ₹ x
The selling price of 18 beds = the cost price of 18 beds + the cost price of 3 beds.
So, the selling price of 18 beds
= the cost price of 21 beds
= ₹16800
The cost price of 21 beds = ₹21x
21x = 16800
x = ₹ 800

25. By selling an item, Madan earned a profit equal to the 1/4th of its cost price. If he would sell it for ₹ 375, then what was the cost price?
(a) ₹312.50 (b) ₹350
(c) ₹300 (d) ₹281.75

RRB JE - 29/05/2019 (Shift-I)

Ans : (c) Let Cost price = ₹ x
According to the question,
Profit $= \frac{x}{4}$
Profit = Selling price - Cost price
$\frac{x}{4} = 375 - x$
$x + \frac{x}{4} = 375$
$\frac{5x}{4} = 375$
$x = 375 \times \frac{4}{5}$
$x = 75 \times 4$
x = ₹ 300
Hence, Cost price ₹300

26. On selling 17 balls at ₹ 720, there is a loss equal to the cost of 5 balls. What is the cost of a ball?
(a) ₹ 40 (b) ₹ 55
(c) ₹ 60 (d) ₹ 45

RRB JE - 02/06/2019 (Shift-II)

Ans. (c) 17 balls → ₹720
According to the question,
There is a loss equal to the cost of 5 balls.
(17–5) balls → ₹720
12 balls → ₹720
1 ball = $\frac{720}{12}$ = ₹60

27. Jack sells a garment for ₹ 1440 and earns 20% profit. What will be the cost price of the garment?
(a) ₹1152 (b) ₹1240
(c) ₹1200 (d) ₹1180

RRB RPF SI -10/01/2019 (Shift-II)

Ans : (c) Selling price = ₹1440
Profit % = 20%
Cost price $= \frac{\text{Selling Price} \times 100}{(100 + \text{Profit\%})}$
$= \frac{1440 \times 100}{100 + 20} = \frac{1440 \times 100}{120}$ = ₹1200

28. On selling a transistor at ₹ 572, a shopkeeper gets a profit equal to 30% the cost price of the transistor. Find the cost price of the transistor.
(a) ₹400 (b) ₹440
(c) ₹340 (d) ₹420

RRB JE - 28/05/2019 (Shift-II)

Ans : (b) Let the cost price of the transistor = ₹ x.
Profit $= x \times \frac{30}{100}$ = ₹ $\frac{3x}{10}$
∵ Selling price = Cost price + Profit
$\therefore 572 = x + \frac{3x}{10}$
$572 = \frac{10x + 3x}{10}$
13x = 5720
x = ₹ 440

29. A seller sells 12 chairs at a profit of 12% and 4 chairs at a loss of 3%. If his total profit is ₹ 1650, the cost price of each chair is:
(a) ₹ 1490 (b) ₹ 1250
(c) ₹ 1100 (d) ₹ 1380

RRB RPF SI -06/01/2019 (Shift-I)

Ans : (b) Let the cost price of each chair = ₹ x.
According to the question,-

Selling price $= 12x \times \frac{(100+12)}{100} + 4x \times \frac{(100-3)}{100}$

$= \frac{3x \times 112}{25} + \frac{97x}{25}$

$= \frac{336x + 97x}{25} = \frac{433x}{25}$

Cost price = 12x + 4x = 16x

Profit = Selling price − Cost price

$1650 = \frac{433x}{25} - 16x$

1650 × 25 = 433x – 400x

$x = \frac{1650 \times 25}{33}$

x = 50 × 25 = 1250

x = ₹1250

30. Rishi sold a pair of shoes at 2,059 including the profit of 16%. What was the cost price of the shoes?

(a) ₹1,800 (b) ₹1,760
(c) ₹1,780 (d) ₹1,775

RRB RPF Constable -19/01/2019 (Shift-II)

Ans : (d) The selling price of shoes = ₹ 2,059
Profit = 16%

$\frac{C.P.}{100} = \frac{S.P.}{Profit + 100}$

The cost price of shoes, $= \frac{2,059 \times 100}{(100+16)}$

$= \frac{2,059 \times 100}{116}$ = ₹1,775

31. 2/3 share of an item sold at a profit of 6% and the rest part was sold at a loss of 3%. If the total profit is ₹ 540, then what was the total cost of the item?

(a) ₹17, 000 (b) ₹18, 000
(c) ₹16, 500 (d) ₹18, 500

RRB Group-D – 24/10/2018 (Shift-I)

Ans : (b) Let the total cost of item = ₹ x.
According to the question,,

$x \times \frac{2}{3}$ selling price $= \frac{2x}{3} \times \frac{106}{100} = \frac{212x}{300}$

Rest part $= x - \frac{2x}{3} = \frac{x}{3}$

∴ The selling price of $\frac{x}{3}$ share $= \frac{x}{3} \times \frac{97}{100} = \frac{97x}{300}$

The selling price of total item $= \left(\frac{212x}{300} + \frac{97x}{300}\right) = \frac{309x}{300}$

Profit $= \frac{309x}{300} - x = 540$

$\frac{9x}{300} = 540$

x = ₹18000

32. Sharad bought two bags for ₹ 900. He sold one of them at a profit of 25% and other at a loss of 25%. If the selling price both bags is equal, then the cost prices of both are respectively:

(a) ₹ 437.5 and ₹ 462.5
(b) ₹ 330 and ₹ 570
(c) ₹ 347.5 and ₹ 552.5
(d) ₹ 337.5 and ₹ 562.5

RRB Group-D – 30/10/2018 (Shift-III)

Ans. (d) : Let the cost price of first bag = ₹ x
And the selling price of the second bag = ₹ (900 – x)
According to the question,,

$x \times \frac{125}{100} = (900 - x)\frac{75}{100}$

5x = 2700 – 3x

8x = 2700

$x = \frac{2700}{8} = 337.5$

So, the cost price of first bag = 337.5
And the cost price of the second bag
= 900–337.5 = 562.5

33. The difference between a loss of 13% and a profit of 15% was ₹ 63. The cost price of the item is:

(a) 225 (b) 207
(c) 198 (d) 243

RRB Group-D – 26/11/2018 (Shift-III)

Ans : (a) Let the cost price of the item = ₹ X.

On selling at 13% loss $= \frac{(100-13) \times X}{100} = \frac{87X}{100}$

On selling at 15% profit $= \frac{(100+15) \times X}{100} = \frac{115X}{100}$

According to the question,,

$\frac{115X}{100} - \frac{87X}{100} = 63$

$\frac{28X}{100} = 63$

$X = \frac{63 \times 100}{28}$ = ₹225

34. On giving a discount on selling price of a chair of 10% and 14%, there is a difference of ₹ 27. Find the cost of the chair.

(a) ₹ 586 (b) ₹ 440
(c) ₹ 675 (d) ₹ 880

RRB Group-D – 08/10/2018 (Shift-II)

Ans : (c) Let the cost of the chair = ₹ x.
According to the question,

$= \frac{x \times 90}{100} - \frac{x \times 86}{100} = 27$

$\Rightarrow \frac{4x}{100} = 27$

$\Rightarrow x = 27 \times 25$

$\Rightarrow$ x = ₹ 675

35. A mobile, when sold at a profit of 6% then ₹ 870 are gained in the comparison to sell it at a loss of 6%. What is the cost price of the mobile phone?

(a) ₹ 6000 (b) ₹ 7000
(c) ₹ 6265 (d) ₹ 7250

RRB Group-D – 22/09/2018 (Shift-I)

Ans : (d) According to the first condition,

$$S.P. = \frac{C.P. \times (100 + Profit\%)}{100}$$

$$= \frac{x \times 106}{100}$$

According to the second condition,

$$S.P. = \frac{C.P. \times (100 - Loss\%)}{100}$$

$$= \frac{x \times 94}{100}$$

Hence, $\frac{106x}{100} = \frac{94x}{100} + 870$

$$\frac{106x}{100} - \frac{94x}{100} = 870$$

$$\frac{12x}{100} = 870$$

$$x = \frac{870 \times 100}{12} = ₹7250$$

36. The difference between a loss of 18% and a profit of 17% was ₹ 63. The cost price of the item is:

(a) ₹189 (b) ₹180
(c) ₹175 (d) ₹198

RRB Group-D – 05/12/2018 (Shift-III)

Ans : (b) According to the question,,

$$\frac{117x}{100} - \frac{82x}{100} = 63$$

$$\frac{35x}{100} = 63$$

$$x = ₹180$$

37. If an item is sold at 13% loss and 14% profit, then the difference between both prices is ₹ 162. What is the cost price of the item?

(a) ₹ 625 (b) ₹ 620
(c) ₹ 600 (d) ₹ 640

RRB Group-D – 03/12/2018 (Shift-II)

Ans : (c) Let the cost price of the item = ₹ x,
According to the question,,

$$x \times \frac{114}{100} - x \times \frac{87}{100} = 162$$

$$\Rightarrow \frac{x}{100}[114 - 87] = 162$$

$$\Rightarrow x = \frac{162 \times 100}{27}$$

$$x = 6 \times 100$$

$$x = 600$$

38. An item was sold at a loss of 12%. If it were sold at much ₹ 49, then there would be a profit of 2%. The cost price of the item is ₹……..

(a) 325 (b) 300
(c) 375 (d) 350

RRB Group-D – 16/11/2018 (Shift-I)

Ans. (d) Let Cost price = ₹ x
On selling at a loss of 12%,

$$\text{Selling price} = \frac{88x}{100}$$

On selling at a profit of 2%,

$$\text{Selling price} = \frac{102x}{100}$$

According to the question,,

$$\frac{102x}{100} - \frac{88x}{100} = 49$$

$$14x = 4900$$

$$x = ₹350$$

39. On selling an item, the difference between 33% loss and 7% profit is ₹ 220. What is the cost price of the item?

(a) ₹ 600 (b) ₹ 525
(c) ₹ 575 (d) ₹ 550

RRB Group-D – 11/12/2018 (Shift-II)

Ans : (d) Let the cost price = ₹ x.
∴ According to the question,,

$$x \times \frac{107}{100} - \frac{x \times 67}{100} = 220$$

$$x \times \frac{40}{100} = 220 \Rightarrow x \times \frac{4}{10} = 220$$

x = ₹ 550

40. Spoorti had sold a pair of shoes at a profit of 17% for ₹ 2,223. What was the cost price of the shoes?

(a) ₹ 1,905 (b) ₹ 1,870
(c) ₹ 1,880 (d) ₹ 1,900

RRB Group-D – 01/10/2018 (Shift-II)

Ans. (d) : Let the cost price = ₹ x.
Profit % = 17%
Selling price = ₹ 2223

$$\therefore \ CP = \frac{SP}{(100 + P\%)} \times 100$$

$$= \frac{2223}{117} \times 100$$

$$= ₹1900$$

41. A person buys two watches for ₹ 480. He sells one of them at 15% loss and the other at 19% profit. After that he comes to know, that he has sold both watches at the same price. Find the cost price of both.

(a) ₹280, 200 (b) ₹270, 190
(c) ₹285, 200 (d) ₹280, 205

RRB Group-D – 18/09/2018 (Shift-III)

Ans. (a) : Let the cost price of the first watch = ₹ x.
The cost price of the second watch = ₹ (480 - x)
According to the question,,

$$x \times \frac{85}{100} = (480 - x) \times \frac{119}{100}$$

$\Rightarrow$ $5x = 480 \times 7 - 7x$

$\Rightarrow$ $12x = 480 \times 7$

x = ₹ 280

And the cost price of the second watch

= 480 – 280 = ₹ 200

42. An item was sold at a loss of 20%, for ₹ 15,000. Find the cost price of the item.

(a) ₹17,750 (b) ₹17,250
(c) ₹18,750 (d) ₹18,250

RRB NTPC 05.04.2016 Shift : 2

Ans : (c) Cost price = $\left(\frac{100}{100 - \text{Loss\%}}\right) \times$ selling price

$$= \left(\frac{100}{100-20}\right) \times 15000$$

$$= \frac{100}{80} \times 15000 = ₹18750$$

43. A shopkeeper earns a profit of 33% even if he gives a discount of 12%. What is the cost price of a chair for shopkeeper, whose face value is ₹ 4740?

(a) ₹3136 (b) ₹4050
(c) ₹3674 (d) ₹3497

RRB NTPC 04.04.2016 Shift : 2

Ans : (a) The face value of the chair = ₹ 4740
The selling price of the chair,

$$= 4740 \times \frac{100-12}{100} = 474 \times \frac{88}{10}$$

Cost price $= 474 \times \frac{88}{10} \times \frac{100}{100+33}$

$$= \frac{474 \times 88 \times 10}{133} = 3136.24 \approx 3136$$

44. A shopkeeper fixed the price of a new item ₹ 1280. If he gets a profit of 20% even after giving a discount of 10%, then find the cost price of the item.

(a) ₹1120 (b) ₹960
(c) ₹1000 (d) ₹940

RRB NTPC 04.04.2016 Shift : 3

Ans : (b) Let the cost price of the item = ₹ x.
Face value = ₹ 1280, Discount = 10%

Selling price of the item = $1280 \times \frac{100-10}{100} = 1280 \times \frac{90}{100}$

According to the question,

$$x = 1280 \times \frac{90}{100} \times \frac{100}{100+20}$$

$$x = 1280 \times \frac{90}{120} \quad x = ₹\ 960$$

45. A shopkeeper fixed ₹ 160, the face value of an item. If he gets a profit of 20%, even after giving a discount of 10% on cost price, then find the cost price of the item.

(a) ₹140 (b) ₹120
(c) ₹150 (d) ₹132

RRB NTPC 02.04.2016 Shift : 1

Ans : (b)

The selling price of the item $= 160 \times \frac{90}{100} = 144$

$\therefore$ Cost price $= \left(\frac{100}{100+20}\right) \times 144 = \frac{100}{120} \times 144 = ₹120$

46. A shopkeeper fixed the value of an item ₹ 320. If he gets a profit of 20%, even after giving a discount of 10% on cost price, then find the cost price of the item.

(a) ₹240 (b) ₹280
(c) ₹300 (d) ₹264

RRB NTPC 02.04.2016 Shift : 2

Ans : (a) Cost price = CP, Face value = MP = 320
Profit = 20%, Discount = 10%

$$\frac{CP \times 120}{100} = \frac{M.P. \times 90}{100}$$

$$\frac{CP \times 120}{100} = \frac{320 \times 90}{100}$$

$$CP = \frac{32 \times 9 \times 10}{12}$$

CP = ₹240

47. Three boxes have 25 packets of 10 pencils each, which were sold for ₹ 8,625, if the profit is 15%, find their cost price.

(a) ₹7,400 (b) ₹7,500
(c) ₹7,600 (d) ₹7,700

RRB NTPC 31.03.2016 Shift : 1

Ans : (b) Cost price $= \left(\frac{100}{100+15}\right) \times 8625$

$$= \frac{100}{115} \times 8625 = ₹7500$$

48. A wholesaler sold a water purifier at a loss of 40%. If the selling price has been increased by ₹125, then wholesaler will get the profit of 10%. What was the cost price of the purifier?

(a) ₹ 250 (b) ₹ 225
(c) ₹ 275 (d) ₹ 300

RRB NTPC 18.04.2016 Shift : 1

Ans : (a) Let the cost price of the water purifier is ₹ x.
According to the question,-

$$x \times \frac{(100-40)}{100} + 125 = \frac{x \times 110}{100}$$

$$\Rightarrow \frac{60x}{100} + 125 = \frac{110x}{100}$$

$$\Rightarrow \frac{50x}{100} = 125$$

$$\Rightarrow \frac{x}{2} = 125$$

$$\Rightarrow x = 125 \times 2 = 250$$

49. When Bablu reduces the selling price of shoes from ₹ 360 to ₹ 345, then he gets 4% much loss. Find the cost price of the shoes.

(a) ₹275 (b) ₹375
(c) ₹425 (d) ₹450

RRB NTPC 16.04.2016 Shift : 1

Ans : (b) According to the question,
4% = 360–345 = ₹15

$$\therefore 100\% = \frac{15}{4} \times 100$$

= ₹375

So, the selling price of the shoes = ₹ 375

50. Sahil sold an item for ₹280 at a loss of 20%. The cost price of the item was :

(a) 336 (b) 340
(c) 350 (d) 1,400

RRB ALP & Tec. (20-08-18 Shift-III)

Ans : (c) Cost price $= \frac{\text{Selling price} \times 100}{(100 - \text{Loss}\%)}$

$$= \frac{280 \times 100}{80} = 350$$

Type - 3

51. An item was sold at a loss of 5%, for ₹ 1235. Find the selling price of the item at the profit of 10%.

(a) ₹1,335 (b) ₹1,380
(c) ₹1,430 (d) ₹1,300

RRB RPF Constable -22/01/2019 (Shift-III)

Ans : (c) $CP = \frac{SP \times 100}{100 - \text{loss}\%}$ (SP = Selling price)

$CP = \frac{SP \times 100}{100 - 5}$ (CP = Cost price)

$$CP = \frac{1235 \times 100}{95} = 1300$$

$$SP = CP \times \frac{100 + \text{Profit}\%}{100}$$

$$SP = CP \times \frac{100 + 10}{100}$$

$$SP = 1300 \times \frac{110}{100} = ₹\ 1430$$

52. K buys a car in ₹ 4.50 lacs and spends ₹ 1.25 lacs on its accessories. He sells the car at a loss of 20%. Find the selling price of the car.

(a) ₹4.00 lacs (b) ₹4.20 lacs
(c) ₹4.40 lacs (d) ₹4.60 lacs

RRB RPF SI -11/01/2019 (Shift-III)

Ans : (d) Total price of the car = 4.50 = 1.25 = ₹ 5.75 lacs

$$\therefore \text{ Selling price} = \frac{100 - \text{Loss}\%}{100} \times \text{Cost price}$$

$$= \left(\frac{100 - 20}{100}\right) \times 5.75$$

$$= \frac{80}{100} \times 5.75 = 4.600 = ₹4.60 \text{ Lacs}$$

53. A cycle bought at ₹ 1400 is sold at a loss of 15%. What is the selling price?

(a) ₹1290 (b) ₹1090
(c) ₹1190 (d) ₹1385

RRB JE - 01/06/2019 (Shift-I)

Ans : (c)
The selling price of the cycle

$$= \text{Cost price} \times \frac{(100 - Loss\%)}{100}$$

$$= 1400 \times \frac{100 - 15}{100}$$

$$= 1400 \times \frac{85}{100}$$

= 14 × 85 = 1190

54. There is loss of 25% on selling an article for ₹2400. What should be the selling price to get the profit of 25%?

(a) ₹2700 (b) ₹4000
(c) ₹3600 (d) ₹4200

RRB JE - 01/06/2019 (Shift-I)

Ans : (b)
The cost price of the article

$$= \text{Selling price} \times \frac{100}{(100 - Loss\%)}$$

$$= 2400 \times \frac{100}{(100 - 25)}$$

$$= 2400 \times \frac{100}{75}$$

= 3200

The selling price of the article

$$= \text{Cost price} \times \frac{(100 + Profit)}{100}$$

$$= 3200 \times \frac{100 + 25}{100}$$

$$= 3200 \times \frac{125}{100}$$

= 32 × 125 = ₹4000

55. A seller gets 22% loss on selling a set of books for ₹ 1,755. What should be the selling price of them for getting 6% profit?

(a) ₹2,375 (b) ₹2,385
(c) ₹2,355 (d) ₹2,365

RRB RPF Constable -20/01/2019 (Shift-II)

Ans. (b)

$$\text{Cost price} = \frac{\text{Selling price}}{(100 \pm \text{Loss/Profit}\%)} \times 100$$

Let Selling price = ₹x

According to the question,,

$$\frac{1755}{100-22} \times 100 = \frac{x}{100+6} \times 100$$

$$\frac{1755}{78} = \frac{x}{106}$$

$$x = \frac{1755 \times 106}{78}$$

$$x = \frac{186030}{78}$$

x = ₹ 2385

56. On selling an old phone at ₹ 6360, Ranjita got 47% less than the cost of buying it a few years ago. At what price should Ranjita sell it to get the benefit of 13%?

(a) ₹13,560 (b) ₹10,550
(c) ₹11,550 (d) ₹ 12,550

RRB Group-D – 05/10/2018 (Shift-I)

Ans. (a) : Let the cost price of the phone is ₹ x.

$$\therefore x \times \frac{(100-47)}{100} = 6360$$

$$\Rightarrow x = \frac{6360 \times 100}{53}$$

= ₹12,000

So, the selling price at 13% profit

$$= \text{Cost price} \times \frac{100+13}{100}$$

$$= 12,000 \times \frac{113}{100} = ₹\ 13,560$$

57. A person bought an item for ₹ 96 and sold it at a profit of 25%, then what was the selling price of the item?

(a) ₹120 (b) ₹125
(c) ₹114 (d) ₹115

RRB Group-D – 19/09/2018 (Shift-III)

Ans. (a) :

$$\text{Profit} = \frac{\text{Selling price}}{\text{Cost price}} \times 100$$

$$125 = \frac{\text{Selling price}}{96} \times 100$$

$$\text{Selling price} = \frac{125 \times 96}{100} = ₹120$$

58. Pavan sold an item at a loss of 12.5%. If he could have sold it for ₹ 56 more, he would have made a profit of 22.5%. What should be the selling price of the item to make a profit of 25%?

(a) ₹ 182 (b) ₹ 190
(c) ₹ 185 (d) ₹ 200

RRB Group-D – 19/09/2018 (Shift-II)

Ans. (d) : Let the cost price of the item is x and the selling price is y. then,

$$12.5 = \frac{(x-y) \times 100}{x}, \text{ and } 22.5 = \frac{(y+56-x) \times 100}{x}$$

$$\text{or, } 22.5 = \frac{(y+56-x) \times 100 \times 12.5}{(x-y) \times 100}$$

$$\Rightarrow \quad 22.5(x-y) = (y-x+56) \times 12.5,$$

$$\Rightarrow \quad 9(x-y) = (y-x+56) \times 5$$

$$\Rightarrow \quad 9x - 9y = 5y - 5x + 280$$

$$\Rightarrow \quad 14x - 14y = 280$$

$$\Rightarrow \quad x - y = 20$$

$$\text{Now, } 12.5 = \frac{20 \times 100}{x} \text{ or } x = \frac{2000}{12.5} = ₹160$$

So, the cost price of the item = ₹ 160

Then, the selling price of the item for 25% profit

$$= \frac{25 \times 160}{100} + 160 = 5 \times 8 + 160 = 40 \times 160 = ₹200$$

59. A person bought an article for ₹ 1975 and sold it at a profit of 12%. What was the selling price of the article?

(a) ₹ 2,212 (b) ₹ 2,192
(c) ₹ 2,222 (d) ₹ 2,202

RRB Group-D – 23/09/2018 (Shift-I)

Ans : (a) The cost price of the article = ₹ 1975

Profit% = 12%

$$\left[\text{S.P.} = \text{C.P.} \times \frac{100 \pm \text{Profit/Loss}}{100}\right] \text{(S.P. = Selling price \& C.P. = Cost price)}$$

$$\text{S.P.} = \frac{1975 \times (100+12)}{100}$$

$$= \frac{1975 \times 112}{100}$$

$$= \frac{221200}{100} = 2212$$

S.P. = ₹2,212

60. On selling a jute bag for ₹48, Ashmit incurs a loss of 20%. In order to make a profit of 20% what should be the selling price of the jute bag?

(a) ₹ 72 (b) ₹ 52
(c) ₹ 56 (d) ₹ 68

RRB Group-D – 18/09/2018 (Shift-II)

Ans. (a) : Let Cost price (C.P.) = x

Selling price (S.P.) = 48, Loss = 20%

$$\because \quad \text{C.P.} \times \frac{80}{100} = 48$$

C.P. = ₹ 60

If the profit is 20%, then

$$\text{C.P.} \times \frac{120}{100} = \text{SP}$$

$$\frac{60 \times 120}{100} = \text{SP}$$

SP = ₹ 72

61. A toy sold at a loss of 16%, was bought at ₹ 925. What was the selling price of the toy?

(a) ₹785 (b) ₹777
(c) ₹775 (d) ₹787

Ans. (b) : Cost price (CP) = 925 Loss = 16%
Selling price (SP) = ?

$$C.P. = \frac{S.P. \times 100}{(100 - Loss\%)}$$

$$925 = \frac{S.P. \times 100}{(100 - 16\%)}$$

$$925 \times 84 = S.P. \times 100$$

$$S.P. = \frac{925 \times 84}{100}$$

Selling Price = ₹ 777

62. An item sold at a profit of 8%, was bought at ₹ 2275. What was the selling price of the item?

(a) ₹2,093 (b) ₹2,443
(c) ₹2,457 (d) ₹2,453

RRB Group-D – 02/11/2018 (Shift-I)

Ans. (c) Cost price (CP) = ₹2275
Profit% = 8%
Selling price (SP) = ?

$$C.P. = \frac{S.P. \times 100}{100 \pm Profit/Loss\%}$$

$$2275 = \frac{S.P. \times 100}{100 + 8}$$

$$S.P. = \frac{2275 \times 108}{100} = \frac{245700}{100}$$

SP = ₹2457

63. A toy was sold at a loss of 16%, and was bought for ₹ 1125. The selling price of the toy was-

(a) ₹ 960 (b) ₹ 945
(c) ₹ 955 (d) ₹ 975

RRB Group-D – 01/10/2018 (Shift-II)

Ans. (b) : Given-
The cost price (CP) of the toy = ₹ 1125
% Loss = 16%
Selling price (SP) = ?

Formula, $C.P. = \frac{S.P.}{(100 - Loss)} \times 100$

$$\Rightarrow \quad 1125 = \frac{SP}{84} \times 100$$

$$\Rightarrow \quad SP = \frac{1125 \times 84}{100}$$

$$SP = \frac{1125 \times 84}{100} = ₹\ 945$$

64. Dev sold a bus at a loss of 15%, for ₹ 2210. to get 15% profit on the same bus, at what price will it have to sell?

(a) ₹2980 (b) ₹3000
(c) ₹ 2970 (d) ₹2990

RRB NTPC 17.01.2017 Shift-3

Ans : (d) The cost price of the bus $= 2210 \times \frac{100}{100 - 15}$

$$= 2210 \times \frac{100}{85} = ₹2600$$

The selling price at the 15% profit

$$= \text{Cost price} \times \frac{100 + 15}{100}$$

$$= 2600 \times \frac{115}{100} = ₹2990$$

65. An article was sold for ₹ 26000 at a discount of 35%. Find the selling price if the discount was 15%.

(a) ₹36,000 (b) ₹40,000
(c) ₹38,000 (d) ₹34,000

RRB NTPC 05.04.2016 Shift : 3

Ans :(d) Let the cost price of the item = ₹ x.

$$\because \text{Selling price} = \text{Cost price} \times \left[1 - \frac{Discount}{100}\right]$$

$$26000 = x \times \left[1 - \frac{35}{100}\right]$$

$$26000 = x \times \frac{65}{100}$$

$$x = \frac{26000 \times 100}{65}$$

x = ₹40000

If the discount is 15%, then selling price

$$= 40000 \times \left[1 - \frac{15}{100}\right] = 40000 \times \frac{85}{100} = ₹34000$$

66. An item is sold at a loss of 20%, for ₹ 2400. What should be the selling price of the item to sell it at a profit of 20%?

(a) ₹3,300 (b) ₹3,600
(c) ₹3,500 (d) ₹3,400

RRB NTPC 31.03.2016 Shift : 1

Ans : (b)

The cost price of the Item $= \left(\frac{100}{100 - 20}\right) \times 2400$

$$= \frac{100}{80} \times 2400 = ₹\ 3000$$

So, for getting 20% profit the selling price of the item

$$= \left(\frac{100 + 20}{100}\right) \times 3000 = 120 \times 30 = ₹3600$$

67. An article was sold at a discount of 20%, for ₹ 2,400. Find the selling price of the article if the discount is 25%.

(a) ₹2,250 (b) ₹2,000
(c) ₹1,800 (d) ₹2,150

RRB NTPC 31.03.2016 Shift : 2

Ans : (a) Let discount = D, The selling price = S.P.

$$\frac{100 - D_1\%}{100 - D_2\%} = \frac{SP_1}{SP_2}$$

$$\frac{100-20}{100-25}=\frac{2400}{SP_2}$$

$$SP_2=\frac{2400\times 75}{80}$$

$SP_2 = ₹2250$

68. A mobile whose selling price is ₹ 7500, is selling at a profit of 50%. A customer take's so me discount from salesman. But the salesman was strictly instructed not to sell the mobile below 35% profit. In this case, how much selling price will the salesman be able to give to the customer?

(a) ₹6000 (b) ₹5000
(c) ₹6750 (d) ₹5500

RRB NTPC 18.04.2016 Shift : 3

Ans : (c) Given–
Selling price = ₹7500
Profit= 50%

$\therefore$ Cost price $=\frac{7500\times 100}{150}=₹5000$

Again, new Profit = 35%

$\therefore$ Selling price $=\frac{5000\times 135}{100}=₹6750$

69. A shopkeeper buys 100 oranges for ` 60. He spends 15% on transportation. What should be the selling price of 100 oranges for getting 20% profit?

(a) ₹72 (b) ₹81.8
(c) ₹82.8 (d) ₹83.8

RRB NTPC 10.04.2016 Shift : 3

Ans : (c) The cost price of 100 oranges including the transportation expenses

$$=60\times\frac{(100+15)}{100}=60\times\frac{115}{100}=₹69$$

So, the selling price of the 100 oranges for 20% profit

$$=69\times\frac{(100+20)}{100}=69\times\frac{120}{100}=₹82.8$$

70. An article was sold at Rs. 920 at a profit of 15%. Find the selling price to get a profit of 20%.

(a) ₹1,000 (b) ₹980
(c) ₹960 (d) ₹940

RRB NTPC 06.04.2016 Shift : 2

Ans : (c) $CP = SP\times\frac{100}{100+\text{Profit\%}}$

$$CP = 920\times\frac{100}{100+15}$$

$$CP = 920\times\frac{100}{115}$$

$CP = 800$

Now the selling price on earning a profit of 20%

$$SP = CP\times\frac{100+\text{Profit\%}}{100}$$

$$=800\times\frac{100+20}{100}=800\times\frac{120}{100}=₹960$$

71. The cost price of 5kg of wheat and 10 kg of lentil were at ₹ 70 and ₹ 80 per kg respectively. On selling, he gained 10% profit on wheat and 20% profit on sugar. What was the total sale price of all items?

(a) ₹1,375 (b) ₹1,345
(c) ₹1,400 (d) ₹1,350

RRB NTPC 27.04.2016 Shift : 2

Ans : (b)
The cost price of 5 kg of wheat = 70 × 5 = 350
The cost price of 10 kg of lentil = 80 × 10 = 800
According to the question,

The selling price of all items $=\frac{350\times 110}{100}+\frac{800\times 120}{100}$

$=385+960=₹1345$

72. If a person bought an item for ₹ 96 and sold it at a profit of 12.5%, the selling price of the item was:

(a) ₹105 (b) ₹110
(c) ₹112 (d) ₹108

RRB ALP & Tec. (31-08-18 Shift-I)

Ans : (d) Selling price $=96\left(1+\frac{12.5}{100}\right)$

$$=96\times\frac{112.5}{100}=96\times 1.125=₹108$$

73. A defective piece of article which costs is ₹1200 is being sold at a loss at 15%. If the price is further reduced by 5%, find the SP?

(a) ₹ 1000 (b) ₹ 969
(c) ₹ 960 (d) ₹ 990

RRB ALP & Tec. (21-08-18 Shift-I)

Ans : (b) The selling price of the article at 15% loss

$$=1200\times\frac{(100-15)}{100}=1200\times\frac{85}{100}$$

$=₹1020$

Again, the selling price of the article after a loss of 5%

$$=1020\times\frac{(100-5)}{100}=1020\times\frac{95}{100}=₹969$$

74. A computer company sells computers of two brands the same day on which it gains the profit of ₹ 15,000. If the profit gained on 1st brand is 35%, then what was the selling price of the 2nd brand?

(a) ₹ 8,750 (b) ₹ 8,000
(c) ₹ 9,750 (d) ₹ 9,000

RRB Group-D – 25/09/2018 (Shift-III)

Ans. (c) : The profit of both brand computers
= ₹ 15,000

The profit gained by 1st brand = 35% of the total profit.

$$= 15000\times\frac{35}{100}=₹\ 5250$$

So the selling price of the 2nd brand = 15000 –5250
= ₹9,750

Type - 4

75. The set of 2 pants and 4 shirts or 1 pant and 6 shirts costs ₹ 5,600. A shopkeeper decides to sell them separately. He sold 10 shirts for ₹ 6,000. Find the loss or profit on each shirt.

(a) Profit ₹1000 (b) Loss ₹1000
(c) Profit ₹100 (d) Loss ₹100

RRB RPF SI -12/01/2019 (Shift-II)

Ans : (d) 2P + 4S = 5600 ……(i) Let S = Shirt
1 P + 6S = 5600……..(ii) P = Pant
On multiplying by 2 in equation (ii)
2P + 4 S = 5600
_2P + _12S = _11200 On subtracting equation (i)
8S = 5600
1 S = 700
∴ So, the cost price of 1 shirt ₹700
And the selling price of 10 shirt = 6000
∴ So, the selling price of 1 shirt $= \frac{6000}{10} =$ ₹600
∴Loss = 700 – 600 = ₹100

76. If an article is sold for ₹ 8000 at a profit of 25%, then find the profit?

(a) ₹1200 (b) ₹1600
(c) ₹1800 (d) ₹1000

RRB JE - 30/05/2019 (Shift-II)

Ans : (b)
The cost price of the article = $8000 \times \frac{100}{125} = 6400$
Hence, Profit = Selling price − Cost price
= 8000 – 6400 = ₹ 1600

77. For selling old articles, a person sold a tea set for ₹ 3540, which was 41% less than the cost price. The seller should have sold the set for ₹ …….. much, to gain 11% profit.

(a) ₹2,460 (b) ₹1,812.60
(c) ₹3,120 (d) ₹2,664.42

RRB Group-D – 28/09/2018 (Shift-I)

Ans : (c)
The selling price of tea set = ₹ 3540 and Loss = 41%
So, the cost price of the tea set $= \left(\frac{100}{100-41} \times 3540\right)$
= ₹6000
Required profit = 11%
So, the selling price of the tea set $= \left(\frac{100+11}{100} \times 6000\right)$
= ₹6660
So, the required amount = 6660 – 3540 = ₹ 3120

78. A low scale occupation earned a profit of ₹ 75,000 in the last year. This year the profit increased by 25%. What is the real profit earned this year?

(a) ₹ 87,750 (b) ₹ 80,750
(c) ₹ 90,750 (d) ₹ 93,750

RRB Group-D – 28/11/2018 (Shift-I)

Ans : (d)
The earned profit on an increase of 25%
$= 75000 \times \frac{(100+25)}{100}$
$\Rightarrow 75000 \times \frac{125}{100} =$ ₹ 93750

79. A company did the sale of ₹ 5,00,000 in a year. In the 3rd year the sale increased by 35% than 1st year. What is the real profit of 3rd year?

(a) ₹ 2,50,000 (b) ₹ 6,75,000
(c) ₹ 6,00,000 (d) ₹ 1,75,000

RRB Group-D – 02/11/2018 (Shift-II)

Ans. (d) The real profit in 3rd year,
$= 5,00,000 \times \frac{35}{100}$
= ₹ 1,75,000

80. An item was sold at a profit of 12.5% for ₹ 2250. What was the amount of profit?

(a) ₹275 (b) ₹250
(c) ₹225 (d) ₹300

RRB NTPC 05.04.2016 Shift-1

Ans : (b) Given-
Selling price = ₹2250
Profit = 12.5%
∴ Cost price $= \frac{\text{Selling Price} \times 100}{(100+12.5)}$
$= \frac{2250 \times 100}{112.5} =$ ₹2000
So, the amount of profit = Selling price – Cost price
= 2250 – 2000 = ₹250

81. If Reena sells 12 mobile phones for ₹ 1,88,160, whose cost price is ₹ 14056 per phone, then how much total profit he earned?

(a) ₹19,488 (b) ₹17,621
(c) ₹21,014 (d) ₹18, 958

RRB NTPC 29.03.2016 Shift : 1

Ans : (a) The selling price of 12 mobiles = ₹ 188160
The total cost price of 12 mobile phones
= ₹12 × 14056 = 168672
Total profit gained by Reena = 188,160 − 168,672
= ₹ 19,488

82. The MRP of a clock is ₹ 4750 and a discount of 12% is given on its sale. If the shopkeeper has bought it for ₹ 3850 then what will be its profit?

(a) ₹240 (b) ₹570
(c) ₹900 (d) ₹330

RRB NTPC 28.03.2016 Shift : 2

Ans : (d) The MRP = ₹ 4750, The cost price = ₹ 3850
The discount % = 12%

Selling price (SP) $= \frac{MP \times (100 - \text{discount}\%)}{100}$

$SP = \frac{4750 \times (100-12)}{100}$

$SP = \frac{4750 \times 88}{100} = 4180$

Profit = SP – CP = 4180 – 3850 = ₹330

83. Jiva bought an item for ₹ 2500 and sold it at 25% much amount and paid ₹ 125 on it. Find the profit of it in ₹?

(a) ₹500 (b) ₹550
(c) ₹475 (d) ₹625

RRB NTPC 18.01.2017 Shift : 2

Ans : (a) Selling price = Cost price $\times \frac{(100 \pm P/L)}{100}$

Selling price $= \frac{2500 \times 125}{100} = 3125$

Profit = Selling price − (Cost price + Tax)

Profit = 3125– (2500+125)

Profit = 3125–2625 = 500

84. A shopkeeper cheats up to 25% by using low-weight in buying and selling fruits, then his total profit percentage is:

(a) 55.25 (b) 56.25
(c) 56.75 (d) 56.5

RRB NTPC 11.04.2016 Shift : 3

Ans : (b) From formula, $x \pm y \pm \frac{xy}{100}$

$= 25 + 25 + \frac{25 \times 25}{100} = 50 + \frac{625}{100} = 50 + 6.25 = 56.25$

Type - 4

85. On selling an item at a profit of 2% and 18%, there is difference of ₹ 3 in the selling price, then the ratio of both selling prices is :

(a) 51 : 59 (b) 51 : 53
(c) 51 : 60 (d) 55 : 59

RRB RPF Constable -20/01/2019 (Shift-II)

Ans : (a) Let the cost price of the item = ₹ x.
The difference of selling prices = ₹ 3.
According to the question,

$x \times \frac{118}{100} - x \times \frac{102}{100} = 3$

$\frac{16x}{100} = 3$

$x = \frac{75}{4}$

The required ratio $= \frac{75}{4} \times \frac{102}{100} : \frac{75}{4} \times \frac{118}{100}$

= 51 : 59

86. The difference of selling price is ₹ 3, on selling an item at a profit of 4% and 10%, the ratio of the selling price of both is:

(a) 52:55 (b) 51:55
(c) 34:35 (d) 55:52

RRB RPF SI -11/01/2019 (Shift-II)

Ans :(a) Let the cost price of the item = ₹ x.
According to the question,

$\frac{x \times 110}{100} - \frac{x \times 104}{100} = 3$

110x − 104x = 300

6x = 300

Cost price = 50

The selling price at a profit of 4% $= \frac{50 \times 104}{100} = 52$

The selling price at a profit of 10% $= \frac{50 \times 110}{100} = 55$

The required ratio = 52 : 55

87. The difference of selling price is ₹ 3, on selling an item at a profit of 4% and 6%, the ratio of the selling price of both is:

(a) 51:52 (b) 52:53
(c) 51:53 (d) 52:55

RRB JE - 24/05/2019 (Shift-III)

Ans : (b) Let the cost price of the item = ₹ x.
The difference of selling prices = ₹ 3
According to the question,

$x \times \frac{106}{100} - x \times \frac{104}{100} = 3$

$\frac{2x}{100} = 3$

x = 150

So, the required ratio $= 150 \times \frac{104}{100} : 150 \times \frac{106}{100} = 52 : 53$

88. If the loss is 15%, then find the ratio of the cost price and the selling price.

(a) 17 : 20 (b) 15 : 17
(c) 10 : 9 (d) 20 : 17

RRB JE - 27/06/2019 (Shift-I)

Ans : (d) Let the cost price of the item = ₹ x.

Loss = 15%

Selling price = Cost price $\times \frac{(100 - \text{Loss})}{100}$

$= x \times \frac{100-15}{100}$

$= x \times \frac{85}{100} = \frac{17x}{20}$

Cost price : Selling price

$x : \frac{17x}{20}$

20 : 17

89. The difference of cost prices of an item is ₹ 3, at profit of 8% and 12%. The ratio of the selling prices of both is:

(a) 27 : 28 (b) 27 : 29
(c) 29 : 31 (d) 27 : 31

RRB NTPC 17.01.2017 Shift-3

Ans : (a) Let the cost price of the item = ₹ x.
The difference of selling prices = ₹ 3
According to the question,

$$x\times\frac{112}{100}-x\times\frac{108}{100}=3$$

$$\frac{4x}{100}=3,\quad x=\frac{300}{4}=75$$

So, the required ratio $=75\times\frac{108}{100}:75\times\frac{112}{100}$

= 108 : 112 = 27 : 28

90. The difference of selling price is ₹ 3, on selling an item at a profit of 8% and 18%, the ratio of the selling price of both is:

(a) 54:59 (b) 54:61
(c) 59:61 (d) 55:59

RRB NTPC 11.04.2016 Shift : 3

Ans : (a) Let the cost price of the item = ₹ x.
The difference of selling prices = ₹ 3
According to the question,

$$x\times\frac{118}{100}-x\times\frac{108}{100}=3$$

$$\frac{10x}{100}=3$$

x = 30

So, the required ratio = $30\times\frac{108}{100}:30\times\frac{118}{100}$ = 54 : 59

91. Mohan earns a profit of 20% on selling a jeans at 15% discount on printed price. The ratio of the cost price to printed price of the jeans is:

(a) 17:24 (b) 17:34
(c) 16:13 (d) 21:23

RRB NTPC 16.04.2016 Shift : 3

Ans : (a) Let the printed price of the jeans is ₹ 100.

According to the question, Cost price $=\frac{100\times85}{120}$

So, the required ratio,

$\Rightarrow \frac{100\times85}{120}:100$

$\Rightarrow 8500:12000 = 17:24$

92. The difference of selling price is ₹ 3, on selling an item at a profit of 8% and 16%, the ratio of the selling price of both is:

(a) 27 : 29 (b) 27 : 31
(c) 29 : 31 (d) 27 : 32

RRB NTPC 07.04.2016 Shift : 1

Ans : (a) Let the cost price of the item = ₹ x.
The difference of selling prices = ₹ 3
According to the question,

$$\frac{x\times116}{100}-\frac{x\times108}{100}=3$$

116x – 108x = 300

8x = 300

$$x=\frac{300}{8}$$

Ratio of Selling price

$$=\frac{300\times108}{8\times100}:\frac{300\times116}{8\times100}=108:116$$

So, the required ratio = 27 : 29

93. The difference of selling price is ₹ 3, on selling an item at a profit of 4% and 8%, the ratio of the selling price of both is:

(a) 25:27 (b) 26:27
(c) 26.:31 (d) 26:29

RRB NTPC 19.01.2017 Shift : 3

Ans : (b) Let the cost price of the item = ₹ x.
The difference of selling prices = ₹ 3
According to the question,

$$x\times\frac{108}{100}-x\times\frac{104}{100}=3$$

$$\frac{4x}{100}=3$$

x = 75

So, the required ratio = $75\times\frac{104}{100}:75\times\frac{108}{100}$ = 26 : 27

94. The difference of selling price is ₹ 3, on selling an item at a profit of 2% and 16%, the ratio of the selling price of both is:

(a) 51:58 (b) 51:53
(c) 57:58 (d) 55:58

RRB NTPC 26.04.2016 Shift : 2

Ans : (a) Let the cost price of the item = ₹ x.
According to the question,

$$\frac{116x}{100}-\frac{102x}{100}=3,\qquad \frac{14x}{100}=3$$

$$x=\frac{150}{7}$$

Selling price $=\frac{150}{7}\times\frac{102}{100}:\frac{150}{7}\times\frac{116}{100}=\frac{51}{7}:\frac{58}{7}$

The required ratio = 51 : 58

95. The difference of selling price is ₹ 3, on selling an item at a profit of 4% and 12%, the ratio of the selling price of both is:

(a) 13:14 (b) 13:15
(c) 12:15 (d) 13:53

RRB NTPC 26.04.2016 Shift : 3

Ans : (a) Let the cost price of the item = ₹ x.
The difference of selling prices = ₹ 3
According to the question,,

$$=\frac{112x}{100}-\frac{104x}{100}=3$$

$\therefore \quad 8x = 300 \qquad x = \frac{300}{8}$

Ratio of Selling price $= \frac{104x}{100} : \frac{112x}{100}$

$= \frac{104}{100} \times \frac{300}{8} : \frac{112}{100} \times \frac{300}{8}$

$= 104 : 112 = 13 : 14$

96. Vishnu spends ₹5000 in buying 12 tables and some chairs. The cost of one table is ₹50 and that of the one chair is ₹40. What is the ratio of the numbers of the chairs to the number of tables purchased ?

(a) 5 : 2 (b) 55 : 6
(c) 5 : 1 (d) 55 : 4

RRB ALP & Tec. (13-08-18 Shift-I)

Ans : (b) Total expenditure = ₹ 5000
A table costs = ₹ 50
The price of 12 tables = ₹ 50×12 = 600
Rest amount =5000–600= ₹ 4400
A chair costs = ₹ 40
The number of purchased chairs $= \frac{4400}{40} = 110$
The required ratio = 110 : 12
= 55 : 6

Type - 4

97. A shopkeeper cheats up to 1% by using low-weight in buying and selling fruits, then his total profit percentage is:

(a) 2.25 (b) 2.01
(c) 2.75 (d) 2.5

RRB RPF Constable -17/01/2019 (Shift-III)

Ans : (b) The shopkeeper cheats up to 1% by using low-weight in buying and selling fruits.

$\therefore$ Total Profit $= 1 + 1 + \frac{1 \times 1}{100}$

$= 2 + 0.01 = 2.01\%$

98. Two identical items are sold for ₹ 200 each, with 10% profit on one but 10 % loss on the other. What is the net percentage loss or profit?

(a) 2% Profit (b) 2% Loss
(c) 1% Profit (d) 1% Loss

RRB JE - 24/05/2019 (Shift-III)

Ans : (d) If two items are sold at the same % loss and profit then,

It will be $\left(\frac{x^2}{100}\right)$.

$\left(\frac{10^2}{100}\right) = \frac{100}{100} = 1\%$ Loss

99. The printed price of an item is 20% more than the cost price and it is sold at a discount of 20%. Find the accurate magnitude of this sale.

(a) No Profit or Loss (b) 10% Profit
(c) 10% Loss (d) 4% Loss

RRB JE - 26/05/2019 (Shift-I)

Ans : (d) Profit/Loss% $= \left(\pm a \pm b \pm \frac{a.b}{100}\right)$

$= 20 - 20 - \frac{20 \times 20}{100} = -4\%$

$\therefore$ Loss % = 4%

Here – sign denote the loss.

100. Ravi sold two bicycles at the same price, but 20% gained on one and lost 20% on the other. Find his profit or loss in the transaction involved.

(a) 10% Profit (b) 20% Profit
(c) 20% Loss (d) 4% Loss

RRB JE - 01/06/2019 (Shift-III)

Ans. (d) If SP of both items are same.

Net Loss% $= \frac{x^2}{100}\% = \frac{(20)^2}{100}\% = 4\%$

101. An item with the cost price ₹ 1000, is sold at a loss of 5% and another item purchased with the earned money is again sold at a profit of 5%. Find the final loss/profit in the deal.

(a) 2.5% Profit (b) ₹ 2.5 Loss
(c) 2% Loss (d) ₹ 2 Profit

RRB JE - 26/06/2019 (Shift-III)

Ans : (b) Profit/Loss % $= a + b + \frac{ab}{100}$

$= 5 - 5 - \frac{25}{100} = 0.25\%$ Loss

So, the required Loss $= 1000 \times \frac{0.25}{100} =$ ₹ 2.5

102. A trader sells a shirt at 6% less amount than the printed price. He fixed the printed price 15% more than the cost price. What is the profit % earned by the trader?

(a) 8.1% (b) 21%
(c) 13.5% (d) 9%

Ans. (a) : According to the question,,

The profit earned by the trader $= x + y + \frac{xy}{100}$

$= -6 + 15 + \frac{-6 \times 15}{100} = 9 + \frac{-90}{100} = 9 - 0.9 = 8.1\%$

103. A horse and a cow were sold for ₹ 12000 each. The horse was sold at 20% profit and the cow at 10% loss. What is the loss/profit on both deals (up to 2 digits of decimal)?

(a) Profit of ₹1000 Profit
(b) Loss of ₹1000
(c) Profit of ₹666.67
(d) No profit and no loss

RRB Paramedical Exam – 21/07/2018 (Shift-II)

Ans : (c) The cost price of the horse

$$= \text{Selling price} \times \frac{100}{100 + \text{Profit}\%}$$

$$= 12000 \times \frac{100}{(100+20)} = ₹10000$$

$$\text{The Cost price of cow} = 12000 \times \frac{100}{(100-10)}$$

$$= 12000 \times \frac{100}{90} = ₹13333.33$$

Hence, Profit = (12000+12000)–(10000+13333.33)
= 24000 – 23333.33 = ₹666.67

104. Bajaj electronics sold two microwave ovens at ₹4,800 each. On one, it gained 20%, and on the other, it lost 20%. Find the gain or loss percentage for the whole transaction?

(a) 5% Loss (b) 4% Profit
(c) 5% Profit (d) 4% Loss

RRB Group-D – 08/10/2018 (Shift-III)

Ans : (d) Profit/Loss % $= \left(\pm a \pm b \pm \frac{a.b}{100}\right)$

$$= \left(+20 - 20 - \frac{20 \times 20}{100}\right)\%$$

$$= (+20 - 20 - 4)\% \quad = -4 \text{ or } 4\% \text{ loss}$$

105. A shopkeeper cheats to the extent of 11% while buying and selling fruit, by using tampered weights. His total gain in percentage is:

(a) 23.25 (b) 23.21
(c) 23.75 (d) 23.5

RRB NTPC 11.04.2016 Shift : 1

Ans : (b) The profit % gained by the shopkeeper

$$= 11 + 11 + \frac{11 \times 11}{100} = 22 + \frac{121}{100}$$

= 22 + 1.21 = 23.21%

106. A shopkeeper cheats up to 7% by using under-weight in buying and selling fruits, then his total profit percentage is:

(a) 14.25 (b) 14.49
(c) 14.75 (d) 14.55

RRB NTPC 07.04.2016 Shift : 2

Ans : (b) Profit % $= \left(2x + \frac{x^2}{100}\right)\%$

$$= 2 \times 7 + \frac{(7)^2}{100}$$

$$= 14 + \frac{49}{100} = 14 + 0.49 = 14.49\%$$

Type - 4

107. A shopkeeper sold 6 radios at a loss of 20%. Find the profit on which he should sell the television, so that his loss is zero. The cost price of the television is thrice to the each radio.

(a) 40% (b) 50%
(c) 30% (d) 60%

RRB RPF Constable -19/01/2019 (Shift-II)

Ans : (a) Let the cost price of a radio = ₹ 100
So, the cost price of the 6 radios = ₹ 600
According to the question,,

$$\text{Selling price} = 600 \times \frac{80}{100} = \boxed{₹\ 480}$$

$$\therefore \text{ Loss} = 600 - 480$$

= ₹ 120

The cost price of a television = 3 × (The cost of a radio)
= 3 × 100
= ₹ 300

Let the required profit % = x %, then

$$300 \times \frac{x}{100} = 120$$

$3x = 120$

$x = 40\%$

108. Pens are bought at the rate ₹ 40 for 8 and sold at ₹ 40 for 6. Find the loss or profit %.

(a) Profit of 40% (b) Profit of 33.33%
(c) Loss of 30% (d) Profit of 20%

RRB JE - 24/05/2019 (Shift-I)

Ans : (b) The cost price of a pen $= \frac{40}{8} = 5$

The selling price of a pen $= \frac{40}{6} = \frac{20}{3}$

$$\therefore \text{ Profit \%} = \frac{\left(\frac{20}{3} - 5\right)}{5} \times 100 = \frac{5}{3 \times 5} \times 100 = 33.33\%$$

109. Find the profit % on the sale of 33 meter of cloth, if the profit in the sale is equal to the selling price of 11 meter of cloth.

(a) 25% (b) 45%
(c) 60% (d) 50%

RRB JE - 26/05/2019 (Shift-I)

Ans : (d) Let the price of 1 meter cloth is ₹ 1,
So, the selling price = ₹ 33
Profit = ₹ 11
So, the cost price = 33 – 11 = ₹ 22

$$\therefore \text{ Profit \%} = \frac{11}{22} \times 100 = 50\%$$

110. A fruit seller bought 100 oranges for ₹ 80. If 20 oranges from them are rotten then, what should be the selling price for each of the remain oranges to gain 25%?

(a) ₹ 1.20 (b) ₹ 1.25
(c) ₹ 1.50 (d) ₹ 1

RRB JE - 27/05/2019 (Shift-III)

Ans : (b) Cost price = 80
The number of rest oranges = 100 – 20 = 80
The required selling price $= \frac{80 \times 125}{100} = 100$
The selling price of one orange $= \frac{100}{80} =$ ₹ 1.25

111. 10 dozen of chocolates are purchased at the rate of ₹ 10 per dozen and sold at the rate of ₹ 2 per piece. If the merchant spends ₹ 50 on fare, find the percentage profit.

(a) 60% (b) 18%
(c) 50% (d) 40%

RRB JE - 26/06/2019 (Shift-I)

Ans : (a) Let the cost price of 1 dozen chocolates = ₹ 10
So, the cost price of 10 dozen of chocolates = 10 × 10 = ₹ 100
And including the rent = 100 + 50 = ₹ 150
The number of chocolates in a dozen = 12
So, the number of chocolates in 10 dozen = 12 × 10 = 120
The selling price of 1 chocolate = ₹ 2
So, the selling price of 120 chocolates = 120 × 2 = ₹ 240
Profit = Selling price − Cost price
= 240 – 150 = 90
Profit % $= \frac{\text{Profit}}{\text{Cost Price}} \times 100 = \frac{90}{150} \times 100 = 6 \times 10 = 60\%$

112. A person buys apples at the rate of ₹ 34 for 8 and sells them at the rate of ₹ 57 for 12. How much apples will he have to be sold to earn a net profit of ₹45?

(a) 90 (b) 150
(c) 100 (d) 135

RRB JE - 02/06/2019 (Shift-III)

Ans : (a) The cost price of 1 apple = 34/8 = ₹ 4.25
The selling price of 1 apple = 57/12 = ₹ 4.75
Profit = Selling price – Cost price
= 4.75 – 4.25 = ₹0.50 or ₹1/2
So, the profit on 1 apple is ₹ 1/2
₹ 1/2 = 1 apple
₹1 = 2 apple
₹45 = 45 × 2 = 90 Apples
Hence, for ₹45 profit to must be sold 90 apple.

113. The cost price of 20 tables is equal to the selling price of 'x' tables. If there is profit of 25% in this deal then, find the value of 'x'.

(a) 18 (b) 16
(c) 25 (d) 15

RRB RPF SI -12/01/2019 (Shift-II)

Ans : (b) 20CP = x SP
$\frac{CP}{SP} = \frac{x}{20}$
$SP = CP \times \frac{125}{100}$
$\frac{CP}{SP} = \frac{4}{5}$
$\frac{x}{20} = \frac{4}{5}$
x = ₹16

114. A seller buys a dozen of pencils for ₹ 25 and sells at the rate of ₹ 12 for 5 pencils. Find the percentage of Profit or loss?

(a) 15% Loss (b) 15.2% Loss
(c) 15.2% Profit (d) 15% Profit

RRB Group-D – 08/10/2018 (Shift-II)

Ans : (c) ∵ The cost price of 12 pencils = ₹ 25
So, the cost price of 1 pencil = ₹ 25/12
∵ The selling price of 5 pencils = ₹ 12
So, the selling price of 1 pencil = ₹ 12/5
So, the required profit % $= \frac{\frac{12}{5} - \frac{25}{12}}{\frac{25}{12}} \times 100$
$= \frac{\frac{144-125}{60}}{\frac{25}{12}} \times 100 = \frac{19}{60} \times \frac{12}{25} \times 100 = 15.2\%$

115. A person bought 3 oranges for ₹ 1 and two oranges for ₹ 1. On which amount will he have sold each dozen to earn a profit of 20%?

(a) ₹ 8 (b) ₹ 18
(c) ₹ 10 (d) ₹ 6

RRB Group-D – 30/10/2018 (Shift-I)

Ans : (d) ∵ the rate of 3 oranges = ₹ 1
The rate of 1 orange = ₹ 1/3
∵ the rate of 2 oranges = ₹ 1
The rate of 1 orange = ₹ 1/2
So, the rate of the mixture of 2 oranges $= \frac{1}{3} + \frac{1}{2} = \frac{5}{6}$
The rate of the mixture of 1 orange $= \frac{5}{12}$
The rate of 1 orange at the profit of 20% $= \frac{5}{12} \times \frac{120}{100}$
The rate of 1 orange $= \frac{1}{2}$₹
So, the rate of 12 oranges at the profit of 20% = ₹ 6

116. A shopkeeper earns the profit equal to the selling price of 2 clocks by selling 12 clocks. Find his profit in percentage.

(a) 16.67% (b) 22%
(c) 20% (d) 25%

RRB Group-D – 10/10/2018 (Shift-III)

Ans : (c) The selling price of 12 clocks
= the cost price of 12 clocks + the selling price of 2 clocks
The selling price of 10 clocks = the cost price of 12 clocks

$= \frac{SP}{CP} = \frac{12}{10}$

Profit $\% = \frac{2}{10} \times 100 = 20\%$

117. A boy buy eggs at the rate of ₹ 16 for per 18 eggs and sells them at the rate of ₹ 20 for per 22 eggs. What is his profit/loss percentage?

(a) $\frac{23}{11}$% Profit (b) $\frac{23}{11}$% Loss
(c) $\frac{25}{11}$% Profit (d) $\frac{78}{11}$% Loss

RRB Group-D – 15/11/2018 (Shift-III)

Ans : (c) The cost price of an egg = $\frac{16}{18}$

The selling price of an egg $= \frac{20}{22}$

Profit $= \frac{20}{22} - \frac{16}{18}$

$= \frac{360-352}{18 \times 22} = \frac{8}{18 \times 22}$

Profit % $= \frac{\frac{8}{18 \times 22}}{\frac{16}{18}} \times 100 = \frac{8 \times 18}{18 \times 22 \times 16} \times 100$

Profit % $= \frac{25}{11}\%$

118. A man bought some oranges at a rate of 3 fruits of 1 rupee and some more oranges at the rate of 2 fruits of 1 rupee. At what price will he has to sell the oranges per dozen to get 20% profit?

(a) ₹ 5 (b) ₹ 4
(c) ₹ 10 (d) ₹ 6

RRB Group-D – 15/10/2018 (Shift-III)

Ans. (d) : by first order,
Cost price of 3 orange = ₹1

$\therefore$ Cost price of 6 orange $= \frac{1}{3} \times 6 =$ ₹ 2

by Second order,
Cost price of 2 oranges = ₹1

$\therefore$ Cost price of 6 oranges $= \frac{1}{2} \times 6 =$ ₹3

Total cost price of 12 oranges = 2 + 3 = ₹ 5
Cost price of 12 oranges to earn a profit fo 20%

$= 5 \times \frac{120}{100} =$ ₹ 6

119. The cost of 2 pencil, 4 pens and & 8 erasers is Rs. 12 and the cost of 8 pens, 10 pencils and 4 erasers is Rs. 36. How much will be the cost of 3 pencils. 3 pens and 3 erasers?

(a) ₹10 (b) ₹15
(c) ₹12 (d) ₹18

RRB NTPC 02.04.2016 Shift : 3

Ans : (c) If the cost of Pensil, Pen and Erasers are x, y and z
According to question

$2x + 4y + 8z = 12$
$10x + 8y + 4z = 36$
$12x + 12y + 12z = 48$
$x + y + z = 4$
$\therefore 3x + 3y + 3z = 4 \times 3 = 12$

120. When 90 chocolates are sold at ₹160 then a chocolate merchant suffers a loss of 20%. In order to earn a profit of 20% how many chocolate should be sale at ₹96?

(a) 45 (b) 36
(c) 54 (d) 28

RRB NTPC 03.04.2016 Shift : 1

Ans : (b) $\because$ Selling price of 90 chocolates = ₹ 160

$\therefore$ Selling price of 1 chocolate $=$ ₹$\frac{160}{90} =$ ₹ $\frac{16}{9}$

Cost price of 1 chocolate $= \frac{16}{9} \times \frac{100}{(100-20)}$

$= \frac{16}{9} \times \frac{100}{80} =$ ₹$\frac{20}{9}$

Selling price of 1 chocolate in order to earn 20% profit.

$= \frac{20}{9} \times \frac{100+20}{100} = \frac{20}{9} \times \frac{120}{100}$

Selling price of 1 chocolate = $\frac{8}{3}$

No. of chocolates in ₹1 $= \frac{3}{8}$

$\therefore$ No. of chocolate ₹96 $= 96 \times \frac{3}{8} = 36$

121. Vikas buy 5 bananas in ₹4 and sells 4 bananas in ₹5. Find his profit%.

(a) 55.56% (b) 53.25%
(c) 45.50% (d) 56.25%

RRB NTPC 16.04.2016 Shift : 3

Ans : (d) $\because$ Cost price of 5 banana = ₹4

$\therefore$ Cost price of 1 banana = $\frac{4}{5}$

Selling price of 4 bananas = 5

Selling price of 1 banana = $\frac{5}{4}$

Profit = Selling price – Cost price

$= \frac{5}{4} - \frac{4}{5} = \frac{25-16}{20}$

Profit = $\frac{9}{20}$

Hence, Profit% $= \frac{9}{20} \times 100 \times \frac{5}{4} = \frac{225}{4}$ = 56.25%

122. A man buy 10 oranges for ₹ 3 and sells 8 for ₹ 3. Calculate his profit.

(a) 20% (b) 25%
(c) 27% (d) 30%

RRB NTPC 27.04.2016 Shift : 3

Ans : (b) Cost price of 1 orange = ₹ $\frac{3}{10}$

selling price of 1 orange = ₹ $\frac{3}{8}$

$$\text{Profit \%} = \left(\frac{\frac{3}{8} - \frac{3}{10}}{\frac{3}{10}} \times 100 \right)\%$$

$$= \left(\frac{\frac{30-24}{80}}{\frac{3}{10}} \times 100 \right)\%$$

$$= \left(\frac{6}{80} \times \frac{10}{3} \times 100 \right)\% = 25\%$$

123. A fruit seller sells mangoes at the rate of ₹ 9 per kg and he gets loses 10%. At what price per kg should he sell in order to earn a profit of 5%.

(a) ₹10 (b) ₹10.5
(c) ₹9.5 (d) ₹11.11

RRB NTPC 28.04.2016 Shift : 2

Ans : (b) Cost price of 1 kg of fruit $= \left(\frac{100}{100-10} \right) \times 9$

= 10

$\therefore$ selling price of fruit to earn 5 % profit

$$= \left(\frac{100 + \text{profit\%}}{100} \right) \times \text{Cost price}$$

$$= \left(\frac{100+5}{100} \right) \times 10 = ₹\ 10.5$$

Type - 8

124. A sells goods to B on 25% profit at ₹300. B sells it at 10% loss to C. In this deal C cost price is equal to what percent of cost price of A?

(a) 97.5% (b) 112.5%
(c) 87.5% (d) 110%

RRB JE - 23/05/2019 (Shift-I)

Ans : (b) Cost price of A $= \frac{300 \times 100}{125} = 240$

Selling price of B $= \frac{300 \times 90}{100} =$ ₹ 270

According to the question, –

$240 \times \frac{x}{100} = 270$

$x = \frac{225}{2} = 112.5\%$

125. Bela is a farmer. She has some acres of land. Last month there was a very good crop, which gave 90% profit on her initial investment (Which was approx 90000). How much approx money is needed by him to invest in each weather?

(a) ₹ 1,00,000 (b) ₹ 3,00,000
(c) ₹ 6,00,000 (d) ₹ 1,50,000

RRB RPF Constable -24/01/2019 (Shift-I)

Ans. (a) : Let the initial investment = ₹ x

$\therefore$ x × 90% = 90000

$x \times \frac{90}{100} = 90000$

x = ₹ 10000

126. In an unstable market an item was bought on different dates at (i) ₹11, (ii) ₹9 (iii) ₹8 and at ₹10 per unit. It was sold at ₹2 more than its cost price on which item there was highest profit gain in percentage.

(a) (i) (b) (ii)
(c) (iii) (d) (iv)

RRB NTPC 19.01.2017 Shift : 2

Ans : (c) On increasing ₹2 in the cost price then selling price–

(i) 13 (ii) 11 (iii) 10 (iv) 12

(i) % Profit $= \frac{2}{13} \times 100 = 15.38\%$

(ii) % Profit $= \frac{2}{11} \times 100 = 18.18\%$

(iii) % Profit $= \frac{2}{10} \times 100 = 20\%$

(iv) % Profit $= \frac{2}{12} \times 100 = 16.66\%$

The highest profit was on item (iii)

127. A weaver sells a saree at ₹150 to a shopkeeper and earns profit of 25%. The shopkeepers sells the same saree to a customer and earns a profit of 30%. If the weaver could sell the saree directly to the customer at ₹180 then what would be his profit % and what would be the customer's profit in ₹?

(a) 50%, ₹ 25 (b) 60%, ₹ 20
(c) 50%, ₹ 15 (d) 40%, ₹ 20

RRB Group-D – 30/10/2018 (Shift-II)

Ans : (c) Cost price of the saree by the weaver (CP)

$$= 150 \times \frac{100}{125} = ₹120$$

Cost price of shopkeeper (CP) = ₹150

Selling price of shopkeeper (SP) and cost price of the customer $= 150 \times \frac{130}{100} = ₹195$

Therefore when the weaver sells directly to the customer

Profit of weaver % $= \frac{180-120}{120} \times 100 = 50\%$

Profit of customer = 195 - 180 = ₹15

128. John buy four old tractors for ₹2 lacs. He spent total of ₹2 lacs in maintenance and repairing. If among the four tractors, he had sold one tractor at ₹1 lacs at first, then in order to learn total 40% profit, what will be the average selling price of the Rest 3 tractors?

(a) ₹ 1.5 lacs (b) ₹ 1.2 lacs
(c) ₹ 2 lacs (d) ₹ 2.3 lacs

RRB NTPC 19.04.2016 Shift : 3

Ans : (c) Total cost price on 4 tractors = 2 + 3 ₹5 lacs

To earn 40% profit, the selling price of the tractor

$$= \frac{5 \times 140}{100} = 7 \text{ lacs}$$

∵ One tractor is sold for ₹1 lacs

∴ Rest 3 tractors, average selling price

$$= \frac{7-1}{3} = ₹\ 2 \text{ lacs}$$

129. A shopkeeper sold two items, one at 25% profit and second at 15% loss and earned profit of 35%. If the price of the item sold at 25% profit is twice the price of item sold at 15% loss, then find the sum of the cost price of both the items.

(a) 100 (b) 400
(c) 300 (d) 200

RRB Group-D – 28/09/2018 (Shift-I)

Ans : (c)

Let the cost price of the item sold at 25% profit be = 2x

Cost of item sold on 15% loss = x

According to the question,

$$\frac{2x \times 125}{100} + \frac{x \times 85}{100} = 3x + 35$$

$$\frac{250x + 85x}{100} = 3x + 35$$

335 x–300x = 3500

35x = 3500

x = 100

Hence, Cost of second item = 2x = 200

Total sum = 100 + 200 = ₹ 300

130. Reema buy a car at ₹75000. She spends ₹10000 on its repairing. Later she sold this car to cheeru at 15% profit. Cheeru sold it to Ritu at 10% profit. What sum of money was spend by Ritu to purchase the car?

(a) ₹1,02,575 (b) ₹1,05,752
(c) ₹1,02,252 (d) ₹1,07,525

RRB Group-D – 03/10/2018 (Shift-II)

Ans : (d) Price of the car bought by Reema = ₹75000

Repairing expense = ₹10000

Total cost = ₹85000

Selling price of car (to cheeru) $= 85000 \times \frac{115}{100} = ₹97750$

Cost price of cheeru = ₹97750

Selling price of cheeru (to Ritu) $= 97750 \times \frac{110}{100}$

= ₹107525

Hence, the sum of money spent by Ritu to buy the car = ₹107525

131. A man buys a land of ₹3 lacs. He sells its 25% part at 25% loss and 40% part at 25% profit. In order to earn a total of 15% profit for how much money should be sell the remaining part of the land.

(a) ₹1,37,500 (b) ₹1,38,750
(c) ₹1,34,500 (d) ₹1,45,000

RRB Group-D – 11/10/2018 (Shift-II)

Ans : (b) Total cost price = ₹3 lacs

To earn total 15% profit, the selling price

$$= 300000 \times \frac{115}{100} = 345000$$

Cost price –I $= 300000 \times \frac{25}{100} = 75000$

Selling price–I $= 75000 \times \frac{75}{100} = 56250$

Cost price–II $= 300000 \times \frac{40}{100} = 120000$

Selling price–II $= 120000 \times \frac{125}{100} = 150000$

Total Selling price = selling price–I + selling price–II

= 56250 + 150000 = 206250

Remainder = 345000 – 206250 = ₹138750

Hence, in order to earn total 15% profit the rest part of the land has to be sold in ₹138750